Reg. No: ……

Name : …….

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEFSQ200 - FOOD PACKAGING TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration : 2 Hours **Maximum Marks: 70**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall attainability

Part A

Multiple Choice Questions

Answer all Questions

Each question carries **1** mark

1. The primary purpose of food packaging is to: [U] / [CO1]
   a). Improve flavour b). Protect food from contamination
   c) .Increase colour d). Reduce cooking time
2. Primary packaging refers to:
   a). Bulk shipping container b). Packaging in direct contact with food [U] / [CO1]
   c). Warehouse packaging d). Decorative cover
3. Packaging classified based on ability to deform includes: [U] / [CO1]
   a). Primary and secondary b). Flexible, semi-rigid and rigid
   c). Tertiary and quaternary d). Consumer and transport

4. Tertiary packaging is mainly used for: [U] / [CO1]
   a). Direct food contact b). Consumer handling
   c). Transportation and storage d). Product display

5. Pilfer-proof closures are mainly intended to: [U] / [CO2]
   a). Enhance colour b). Prevent tampering
   c). Reduce cost d). Improve flexibility

6. PET bottles are widely used for packaging: [U] / [CO2]
   a). Carbonated beverages b). Milk
   c). Sauces d). All of the above

7. Tinplate used in food cans is: [U] / [CO2]
   a). Zinc coated steel b). Aluminium coated steel
   c). Tin coated steel d). Chromium coated steel

8. Aluminium foil is valued in food packaging mainly for its: [U] / [CO2]
   a). Elasticity b). Transparency
   c). Barrier properties d). Porosity

9. Corrugated fibreboard is commonly used for: [U] / [CO2]
   a). Aseptic packaging b). Direct food contact
   c). Transport packaging d). Liquid food packaging

10. Milk pouches are commonly made from: [U] / [CO2]
    a). PVC b). LDPE
    c). PET d). Nylon

11. Laminates are used in food packaging to: [U] / [CO2]
    a). Reduce shelf life b). Add glossy appearance to package
    c). Increase brittleness d). Improve barrier properties

12. Protective lacquers in metal cans are applied to: [U] / [CO2]
    a). Improve appearance b). Prevent corrosion and contamination
    c). Increase flexibility d). Reduce thickness

13. _____ Authority regulates food packaging standards in India. [A] / [CO3]
    a). WHO b). ISO
    c). FSSAI d). FDA

14. A food manufacturer adopts modified atmosphere packaging (MAP) mainly to: [A] / [CO3]

a). Reduce microbial activity b). Extend shelf life by controlling internal gases

c). Reduce temperature d). Increase package weight

15. While selecting packaging material for acidic foods like fruit juices, which regulation must be primarily considered to prevent metal contamination? [A] / [CO3]

a). Labelling regulation b). Advertisement guidelines

c). Transport standards d). Migration limits under food contact material regulations

16. A food package displaying incorrect mandatory information violates which aspect of packaging regulation? [A] / [CO3]

a). Labelling & legal compliance b). Structural integrity

c). Transportation safety d). Consumer trust

17. Failure of package integrity is most likely to result in: [An] / [CO4]

a). Improved shelf life b). Microbial contamination

c). Better appearance d). Reduced material cost

18. Package integrity testing mainly assesses: [An] / [CO4]

a). Label design b). Seal strength and leakage

c). Product flavour d). Colour stability

19. The test used to evaluate the mechanical strength of packaging materials is _____. [An] / [CO4]

a). Burst test b). WVTR & GTR

c). Microbial test d). Chemical test

20. Performance assessment of packaging materials focuses on: [An] / [CO4]

a). Marketing value b). Functional efficiency during storage and distribution

c). Brand image d). Consumer preference

[1×20=20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

21. Explain the importance and scope of food packaging.	[U] / [CO1]
22. Describe the types of packaging based on contact and package levels.	[U] / [CO1]
23. Discuss the properties, advantages, and disadvantages of glass packaging.	[U] / [CO2]
24. Explain on tinplate and aluminium containers.	[U] / [CO2]
25. Describe the various types of smart food packaging used.	[A] / [CO3]
26. Explain how packaging laws and regulations are applied in the food industry.	[A] / [CO3]
27. Detail the transport worthiness tests used for food packaging testing.	[An] / [CO4]
28. Describe methods used to evaluate the quality of food packaging materials.	[An] / [CO4]

[5×6=30]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Discuss in detail the functions of food packaging and classification based on materials, use, and package levels.	[U] / [CO1]
30. Explain plastic packaging materials with respect to classification, properties, advantages, and limitations.	[U] / [CO2]
31. Describe advanced food packaging technologies and explain the application of packaging regulations in food industries.	[A] / [CO3]
32. Analyse the quality and integrity of food packaging materials using appropriate testing methods and performance assessments.	[An] / [CO4]

[2×10=20]